

AI SKIN INTELLIGENCE

& Personalized Skincare Platform

Project Documentation

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1. Objective

Build an AI-powered skincare intelligence platform that creates personalized skincare routines by analyzing a user's skin profile, lifestyle habits, sleep patterns, environmental exposure, and specific skin concerns.

The platform helps users improve skin health through AI-driven skin assessments, personalized skincare recommendations, ingredient analysis, product recommendations, routine planning, and skin progress tracking.

The solution is designed for skincare consumers, dermatology clinics, wellness platforms, beauty brands, and skincare consultants.

Key Outcomes Achieved

- Developed and deployed an AI-powered skincare intelligence platform
- Implemented secure authentication and role-based access control (User, Consultant, Dermatologist, Administrator)
- Built skin profile assessment and skincare recommendation workflows
- Developed personalized skincare routine generation systems
- Implemented ingredient intelligence and product recommendation engines
- Built skin health scoring and progress tracking modules
- Developed dedicated dashboards for every role with live analytics
- Deployed the platform with Dockerized services on a cloud host

2. System Architecture

The platform follows a layered microservices architecture, separating client access, API routing, business logic, AI/ML intelligence, and data storage. This separation keeps each concern independently scalable and testable.

Layer	Responsibility
Clients / Users	Web app, mobile app, dashboards, visualizations and reports for User, Skincare Consultant, Dermatologist and Administrator roles
API Gateway (FastAPI)	Routing, authentication, rate limiting, request validation, load balancing, CORS
Microservices Layer	User, Skin Profile, Assessment, Routine Planner, Ingredient Intelligence, Product Recommendation, Skin Health Scoring, Progress Tracking, Notification, Analytics, Report and Admin services
AI / ML Intelligence Engine	Skin type classification, concern detection & severity, recommendation & ingredient-suitability models, progress prediction, NLP for product understanding
Data Layer	PostgreSQL (relational), MongoDB (documents), Elasticsearch (search), Vector DB / FAISS (embeddings), Redis (cache), file storage, monitoring & backup
External Services	Skincare product & ingredient databases, dermatology knowledge bases, research publications, weather/UV APIs, payment gateway, cloud storage

Technology Stack

Category	Tools & Frameworks
Backend	Python, FastAPI
Frontend	JavaScript, React.js, Next.js, Tailwind CSS
Primary / Secondary DB	PostgreSQL, MongoDB
AI & Machine Learning	Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy
Recommendation & Analytics	XGBoost, LightGBM, custom recommendation models
Visualization	Chart.js, Plotly
Auth & Security	JWT authentication, OAuth2, role-based access control
Cloud & DevOps	Docker, Docker Compose, AWS / Azure, GitHub Actions
Dev Tools	VS Code, Git & GitHub, Postman

3. Core Modules

3.1 User Authentication & Role-Based Access

User registration and login, JWT authentication, OAuth2 login, role-based access control and profile management, across four roles: User, Skincare Consultant, Dermatologist and Administrator.

3.2 Skin Profile Management

Captures skin type, age group, skin concerns, allergies, sensitivities, lifestyle habits, sleep quality and water intake, feeding lifestyle, sleep-pattern, hydration and environmental-exposure tracking.

3.3 Skin Assessment Engine

Identifies and prioritizes skin concerns (acne, hyperpigmentation, dark spots, dryness, oiliness, sensitivity, wrinkles, fine lines, redness, uneven tone), evaluates overall skin health and scores condition severity and risk factors.

3.4 Personalized Routine Generator

Generates morning, evening, weekly and seasonal routines across cleansing, exfoliation, treatment, moisturizing, sun protection and night-care categories, with adaptive updates as skin data changes.

3.5 Ingredient Intelligence Module

Analyzes ingredient suitability and interactions, detects allergy risk, and educates users on key actives such as retinoids, niacinamide, vitamin C, hyaluronic acid, salicylic acid, ceramides, peptides and AHAs/BHAs.

3.6 Product Recommendation Engine

Delivers personalized, suitability-scored product recommendations with comparisons, alternatives and budget-based suggestions across face wash, moisturizer, sunscreen, serum, toner, treatment products and masks.

3.7 Progress Tracking, Notifications & Reports

Tracks routine adherence and improvement over time with before/after comparisons and trend analysis; sends routine, hydration, sleep and replenishment reminders; and exports skin, routine and recommendation reports as PDF or Excel.

4. Skin Health Scoring Model

Each user's overall Skin Health Score is a weighted composite of five factors:

Factor	Weight
Skin Condition Assessment	35%
Lifestyle Habits	20%
Routine Consistency	20%
Sleep Quality	15%
Hydration Level	10%

5. Consumer Experience — User Role

A registered user signs in to a personalized dashboard that shows their live skin health score, detected skin type, active routine steps, unread notifications, and top skin concerns.

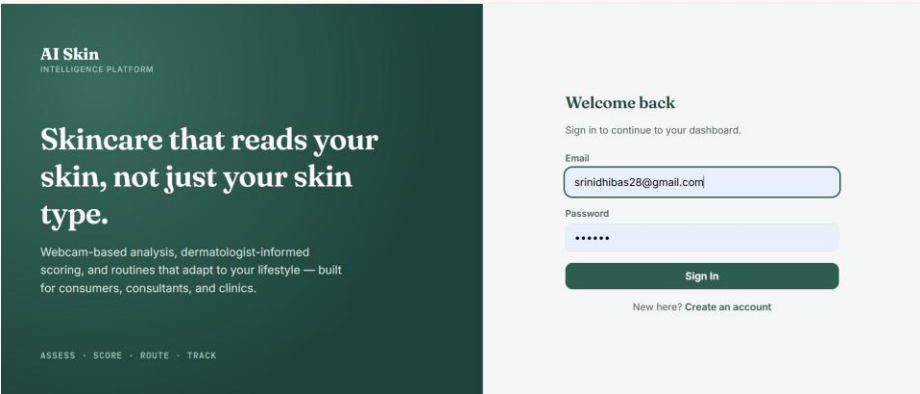


Fig. 1 — Sign-in screen for the User role

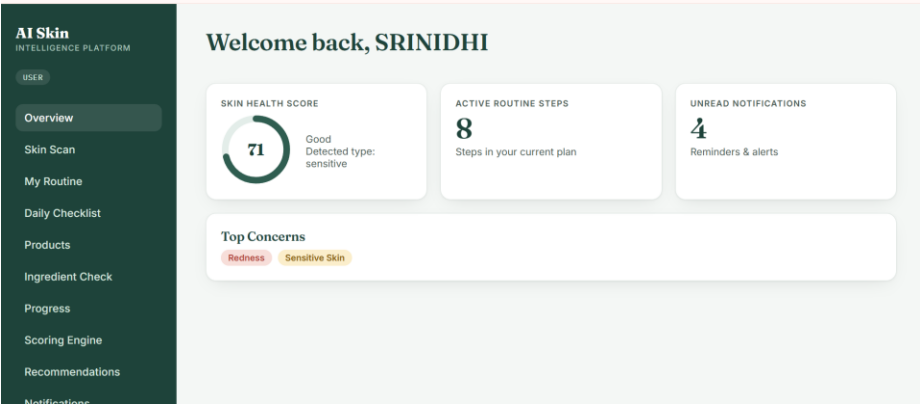


Fig. 2 — User Overview: skin health score 71 (Good, sensitive type), 8 routine steps, 4 notifications, top concerns Redness / Sensitive Skin

6. Clinical & Consultant Views

6.1 Dermatologist Dashboard

Dermatologists sign in to view patient insights, high-risk flags, and detailed skin condition reports, and can link new patients to their caseload by user ID.

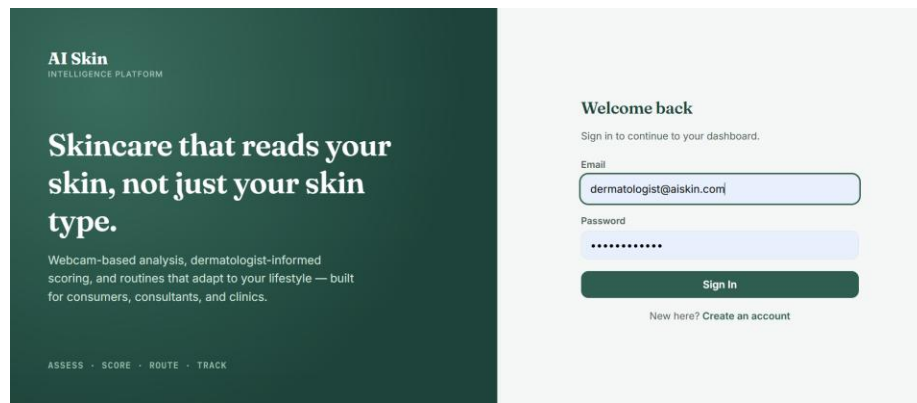


Fig. 3 — Sign-in screen for the Dermatologist role

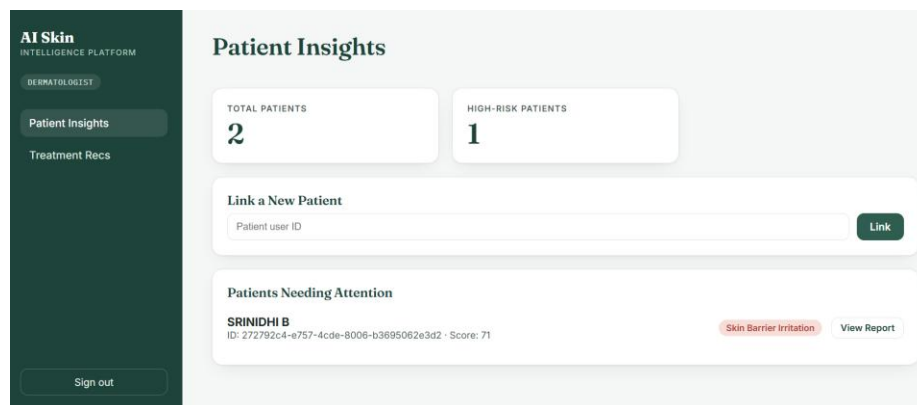


Fig. 4 — Patient Insights: 2 total patients, 1 flagged high-risk (Srinidhi B — Skin Barrier Irritation, score 71)

6.2 Skincare Consultant Dashboard

Consultants manage a client roster, review each client's latest skin score and status, and open detailed reports or recommendation records for any linked client.

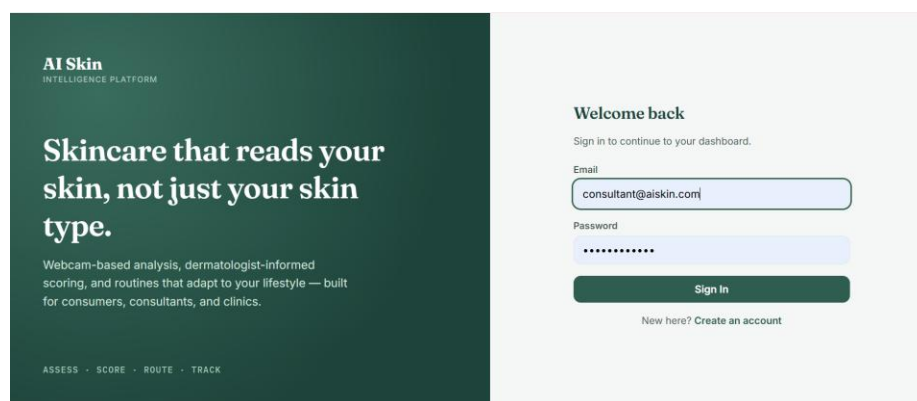


Fig. 5 — Sign-in screen for the Skincare Consultant role

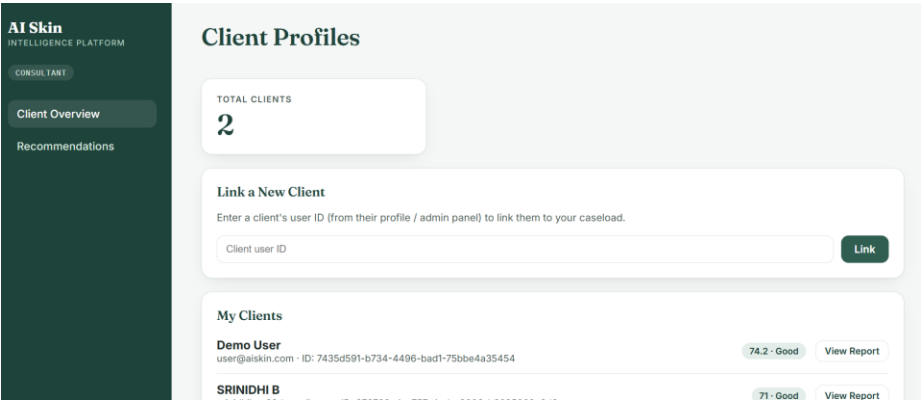


Fig. 6 — Client Profiles: 2 linked clients (Demo User — 74.2 Good, Srinidhi B — 71 Good)

7. Platform Administration

Administrators oversee the full user base across all roles, with the ability to activate or deactivate any account, alongside platform analytics, recommendation monitoring and system reports.

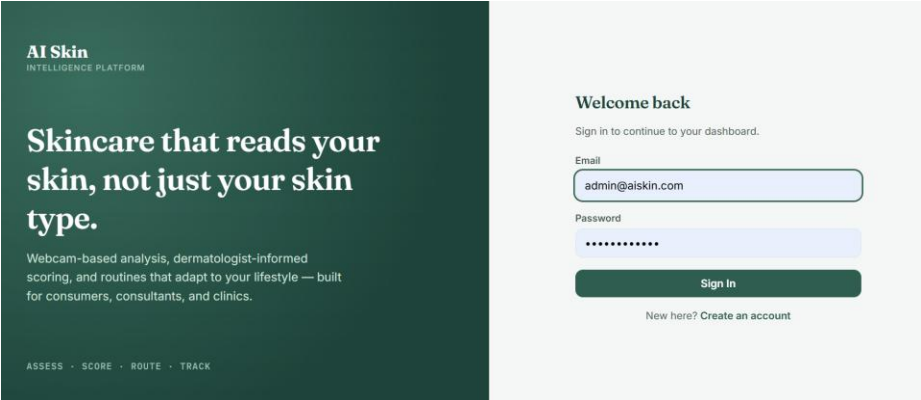


Fig. 7 — Sign-in screen for the Administrator role

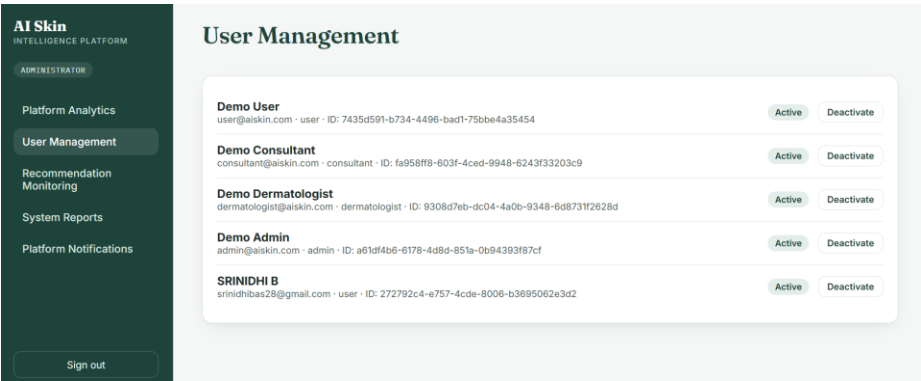


Fig. 8 — User Management: all five active accounts across User, Consultant, Dermatologist and Admin roles

8. Development Milestones

Milestone	Deliverables
Week 1–2 Initialization & Core Setup	System architecture & DB schema, UI wireframes, frontend/backend setup, authentication and role-based access, skin profile & lifestyle tracking
Week 3–4 Assessment & Routine Generation	Skin assessment engine, concern analysis, personalized routine generation, skin scoring workflows

Week 5–6 Product Intelligence & Progress	Ingredient intelligence engine, product recommendation workflows, progress tracking, analytics, user dashboards
Week 7–8 Analytics, Testing & Deployment	Executive dashboards, reports & visualizations, testing & validation, Docker deployment, final documentation

9. Evaluation Criteria

- Milestone 1 (Week 2): Project initialization, authentication, skin profile management and lifestyle tracking all functional
- Milestone 2 (Week 4): Skin assessment engine, personalized routine generation and skin scoring implemented
- Milestone 3 (Week 6): Product recommendation engine, progress tracking and analytics workflows completed
- Milestone 4 (Week 8): Fully deployed frontend and backend with dashboards, reporting and an end-to-end demonstrable workflow

Performance Metrics Tracked

Category	Metrics
Skin Assessment	Concern classification accuracy, skin scoring consistency, recommendation relevance
Product Recommendation	Suitability accuracy, recommendation precision, user satisfaction score
Progress Tracking	Routine adherence accuracy, improvement tracking quality, user engagement rate
System Performance	API response time, dashboard loading speed, recommendation latency, concurrent user handling

10. Conclusion

The AI Skin Intelligence & Personalized Skincare Platform delivers a complete, role-based skincare intelligence workflow — from webcam-based skin assessment through personalized routines, ingredient and product recommendations, to longitudinal progress tracking. All four dashboards (User, Consultant, Dermatologist, Administrator) are live and functional, authentication and account management work end-to-end, and the platform is deployed and reachable via a hosted link, fulfilling the project's original objective.